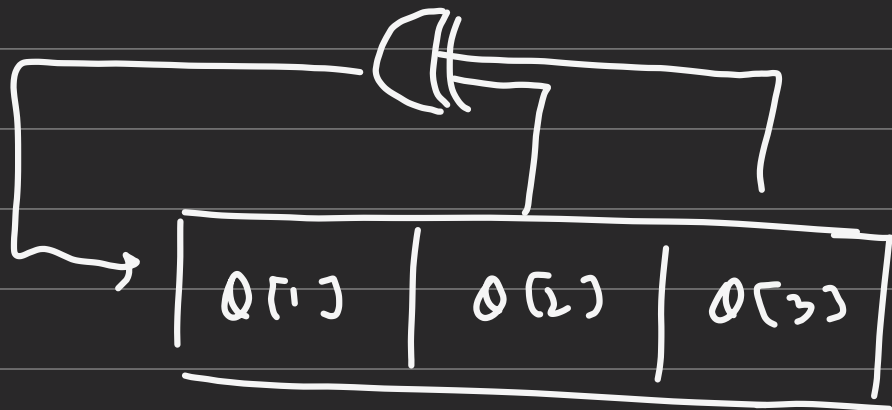


Linear feedback shift Register (LFSR)

↳ generate random no.

↳ used in cryptography

i.e



3 Tap register

```
module LFSR#(—— N=3)
```

```
<———
```

```
output (1:N)Q;
```

```
// state reg. logic
```

```
reg [1:N] Q-reg, Q-next;  
wire tap;
```

```
always @ (posedge clk, ——)
```

```
begin
```

0 → (—)
Q-reg ≤ 'd1;

case 1 to N else
glitches if started Q-reg ≤ Q-next;
from 0 end.

// next state logic

always @ (taps, Q-reg)

Q-next = {taps, Q-reg [1:N-1]};

// output logic

assign Q = Q-reg

assign taps = Q-reg [3] ^ Q-reg [2]